Practice M4: Network. Software. Services (openSUSE)

* NOTE: Some of the exercises in this practice guide are unsuitable for execution in **WSL** or **Docker** environments. A virtual infrastructure should be used instead.

For this practice we will use an **openSUSE Leap 16.x** VM. Alternatively, you could use a different version or distribution like **openSUSE Leap 15.x**, or **openSUSE Tumbleweed.**

All commands that we are going to use in this practice will be shown in combination with a prompt. This way, it will be easier for us to understand which user, and at which station we are working.

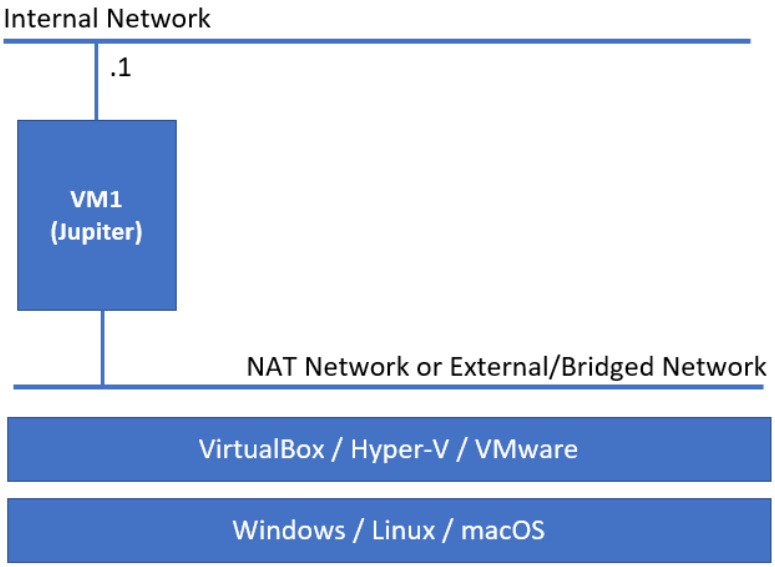
Please note that for each practice we start with a fresh machine or set of machines.

Part 1: Network

Preparation

For this part we will use an **openSUSE Leap 16.x** VM with two network adapters – one in **NAT** mode (or **Bridged** if you prefer), and another in **Internal Network** (set the name to **MyNet**, for example).

For the next steps we will assume that we have imported a new machine and configured it according to the above requirements. In general, this is what we should have:



*Please note that during the import every adapter for which a new MAC address is generated may not be recognized correctly (initially will be in down state without a valid configuration) by the network stack (if **wicked** is in use) and additional steps (described below) have to be taken.*

General

No matter what distribution we use, there is a set of commands that are available. For example, for all distributions that adopted **systemd**, we can change the host's name with:

```
lsauser@opensuse:~> hostnamectl
   Static hostname: opensuse.lsa.lab
   ...
lsauser@opensuse:~> sudo hostnamectl set-hostname jupiter.lsa.lab
lsauser@opensuse:~>
```

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*Note that you can substitute **set-hostname** with **hostname** and the result will be the same. In fact, exactly **hostname** can be seen as a sub-command in the man page about **hostnamectl***

We can add also the so-called pretty name with:

```
lsauser@opensuse:~> sudo hostnamectl set-hostname --pretty 'Jupiter Server'
lsauser@opensuse:~>
```

Our prompt stays the same. Let's check some of the related configuration files:

```
lsauser@opensuse:~> cat /etc/hostname
jupiter.lsa.lab
lsauser@opensuse:~> cat /etc/machine-info
PRETTY_HOSTNAME="Jupiter Server"
lsauser@opensuse:~>
```

The second one may exist or may not exist.

In order for the changes to be reflected in the prompt, we **must close the session** and open a new one.

Now that we are back in, let's ask for the network links.

Let's assume the worst-case scenario – new MAC addresses generated during the import and second network adapted added before the first start. If your situation is different, then please skip some steps that are irrelevant.

If we ask for the network links, we will see something like:

```
lsauser@jupiter:~> ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth2: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:00:27:18:d1:aa brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:00:27:4c:d3:b4 brd ff:ff:ff:ff:ff:ff
lsauser@jupiter:~>
```

As we can see, all adapters are **in a down** state, which means that they are not connected. We cannot make an SSH connection to our machine even if the SSH service is installed and running.

Depending on how the machine was imported and what the active network management solution is, the situation with you may vary. If there is no **eth0** (or **enp0s3**) interface, you must continue with the following steps (**Repair network interfaces**). Otherwise, skip to the **Playing with network interfaces** section.

Repair network interfaces (if needed)

We can use even shorter commands by skipping characters, as long as the given arguments can be translated into a valid combination. For example, the above command can be written like **ip l sh**, or even shorter, if we know that the default action is **show**, it can become **ip l**.

Now, let's bring both interfaces up:

```
lsauser@jupiter:~> sudo ip link set dev eth1 up
lsauser@jupiter:~> sudo ip link set dev eth2 up
```

We can see the current IP address with:

```
lsauser@jupiter:~> ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
```

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```
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:18:d1:aa brd ff:ff:ff:ff:ff:ff
    inet6 fe80::a00:27ff:fe18:d1aa/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:4c:d3:b4 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::a00:27ff:fe4c:d3b4/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
lsauser@jupiter:~>
```

If neither of the network adapters is assigned a valid IP, we can either set manually one or work out the configuration files. Let's first go with the second option.

Create two copies of the existing configuration file and delete the original:

```
lsauser@jupiter:~> sudo cp /etc/sysconfig/network/ifcfg-eth0 /etc/sysconfig/network/ifcfg-eth1
lsauser@jupiter:~> sudo cp /etc/sysconfig/network/ifcfg-eth0 /etc/sysconfig/network/ifcfg-eth2
lsauser@jupiter:~> sudo rm /etc/sysconfig/network/ifcfg-eth0
```

Alternatively, delete the following:

```
lsauser@jupiter:~> sudo rm /etc/udev/rules.d/70-persistent-net.rules
lsauser@jupiter:~>
```

This will remove the remembered interface which had the old MAC address, and which is not available now, after we re-generated it during the import.

In either case, **restart** the system.

*The main difference between the two methods is that in the first, we will end up with **eth2** (in NAT) and **eth1** (in internal network). And in the second approach, the names will be **eth0** (in NAT) and **eth1** (in internal network). So, depending on what you did, adjust the following sections accordingly.*

Playing with network interfaces

If yours are with different indexes and names (**eth0** and **eth1** instead of **enp0s3** and **enp0s8**), you must adjust the commands that follow. Please note that if you came directly to this section, you may need to bring up the second interface.

Once the system has booted, let's login and check the situation with the IP addresses:

```
lsauser@jupiter:~> ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:df:24:6d brd ff:ff:ff:ff:ff:ff
```

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```
    altname enx080027df246d
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 84787sec preferred_lft 84787sec
    inet6 fd17:625c:f037:2:31c8:fc86:b1ae:118e/64 scope global temporary dynamic
       valid_lft 86150sec preferred_lft 14150sec
    inet6 fd17:625c:f037:2:eba5:1db8:33e3:114e/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86150sec preferred_lft 14150sec
    inet6 fe80::6e83:bb01:b686:f3c4/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:60:17:c2 brd ff:ff:ff:ff:ff:ff
    altname enx0800276017c2
    inet6 fe80::2d7:36cc:b16a:9e81/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
lsauser@jupiter:~>
```

As we can see, our **enp0s8** (or **eth1**) adapter still does not have any address assigned. We can add one manually with:

```
lsauser@jupiter:~> sudo ip address add 192.168.200.1/24 dev enp0s8
lsauser@jupiter:~>
lsauser@jupiter:~> ip address show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:60:17:c2 brd ff:ff:ff:ff:ff:ff
    altname enx0800276017c2
    inet 192.168.200.1/24 scope global enp0s8
       valid_lft forever preferred_lft forever
lsauser@jupiter:~>
```

Now, let's try this:

```
lsauser@jupiter:~> ping -c 3 -q 192.168.200.1
PING 192.168.200.1 (192.168.200.1) 56(84) bytes of data.

--- 192.168.200.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 0.040/0.043/0.045/0.002 ms
lsauser@jupiter:~>
```

Without further action, the changes, related to the manually set static address, will be lost on system restart. We don't want to make them permanent, at least not now, so let's tweak the command to delete the address:

```
lsauser@jupiter:~> sudo ip address del 192.168.200.1/24 dev enp0s8
lsauser@jupiter:~>
lsauser@jupiter:~> ip address show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:60:17:c2 brd ff:ff:ff:ff:ff:ff
    altname enx0800276017c.
lsauser@jupiter:~>
```

With the same tool, we can manage the routing table:

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```
lsauser@jupiter:~> ip route show
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
```

Check Which One is Used

openSUSE can run either one of the two network management software solutions - **wicked** and **Network Manager**. Normally you would see one of them up and running. So, how can we check which one is running? There are multiple ways of doing this check.

Option 1 is to execute the following command:

```
lsauser@jupiter:~> systemctl show -p Id network.service
Id=NetworkManager.service
lsauser@jupiter:~>
```

Option 2 is to check which service is installed and running by executing known commands:

```
lsauser@jupiter:~> systemctl status wickedd
● wickedd.service - wicked network management service daemon
   Loaded: loaded (/usr/lib/systemd/system/wickedd.service; indirect; vendor preset: disabled)
   Active: inactive (dead)
lsauser@jupiter:~> systemctl status NetworkManager
● NetworkManager.service - Network Manager
   Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; vendor preset: disabled)
  Drop-In: /usr/lib/systemd/system/NetworkManager.service.d
           └─NetworkManager-ovs.conf
   Active: active (running) since Sun 2019-03-03 16:39:28 EET; 15h ago
...
lsauser@jupiter:~>
```

Option 3 is to use the all-in-one interactive system management utility – **YaST**. It can be started with:

```
lsauser@jupiter:~> sudo yast2
```

Then you must navigate to **System > Network Settings**

Note that YaST is not installed by default on openSUSE Leap 16.0.

How to Do the Switch

<u>This is applicable only to versions of openSUSE Leap 15.x and lower.</u>

You are not obliged to do it. The following steps are included for completeness only. Feel free to skip them.

First, you should make sure that you have them both installed and one of them in use.

For NetworkManager this is done via:

```
lsauser@jupiter:~> sudo zypper install NetworkManager
```

And for wicked via:

```
lsauser@jupiter:~> sudo zypper install wicked
```

Easiest option is to use **YaST** and change the active service from the **System > Network Settings**

Alternative approach is to execute a set of commands. If want to switch from **Network Manager** to **wicked**, we must execute:

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```
lsauser@jupiter:~> sudo systemctl stop network.service
...
lsauser@jupiter:~> sudo systemctl disable NetworkManager.service
...
lsauser@jupiter:~> sudo systemctl enable wicked.service
...
lsauser@jupiter:~> sudo systemctl start network.service
```

In a similar fashion we can switch from **wicked** to **Network Manager**:

```
lsauser@jupiter:~> sudo systemctl stop network.service
...
lsauser@jupiter:~> sudo systemctl disable wicked.service
...
lsauser@jupiter:~> sudo systemctl enable NetworkManager.service
...
lsauser@jupiter:~> sudo systemctl start network.service
```

Switching between the two stacks can cause connectivity issues and may require additional configuration.

When switching from **Network Manager** to **wicked** you may want to save and convert the existing configuration to the new service. For this purpose, you must execute:

```
lsauser@jupiter:~> cd /etc/wicked/ifconfig
lsauser@jupiter:/etc/wicked/ifconfig> sudo wicked show-config compat: | sudo tee all.xml
...
lsauser@jupiter:~> cd /etc/sysconfig/network ; sudo mkdir save
lsauser@jupiter:/etc/sysconfig/network> mv ifcfg-* save/
```

You may still need to adjust the configuration. **YaST** can help you and save you time.

Should you need to re-read the network interface configurations, execute:

```
lsauser@jupiter:~> sudo systemctl restart network
```

Both **xml** and **ifcfg** configurations can be mixed, but it is recommended to stick to one of them, it will make things easier and clearer.

Wicked

If your installation is <u>not using wicked</u>, feel free to skip this section.

We can see the status of all network adapters with:

```
lsauser@jupiter:~> sudo wicked show all
lo               up
      link:      #1, state up
      type:      loopback
      config:    compat:suse:/etc/sysconfig/network/ifcfg-lo
      leases:    ipv4 static granted
      leases:    ipv6 static granted
      addr:      ipv4 127.0.0.1/8 [static]
      addr:      ipv6 ::1/128 [static]

eth2             up
```

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```
      link:     #2, state up, mtu 1500
      type:     ethernet, hwaddr 08:00:27:18:d1:aa
      config:   compat:suse:/etc/sysconfig/network/ifcfg-eth2
      leases:   ipv4 dhcp granted
      leases:   ipv6 dhcp requesting
      addr:     ipv4 10.0.2.15/24 [dhcp]
      route:    ipv4 default via 10.0.2.2 proto dhcp

eth1            setup-in-progress
      link:     #3, state up, mtu 1500
      type:     ethernet, hwaddr 08:00:27:4c:d3:b4
      config:   compat:suse:/etc/sysconfig/network/ifcfg-eth1
      leases:   ipv4 dhcp requesting
      leases:   ipv6 dhcp requesting
lsauser@jupiter:~>
```

From the above output, we can see that we have two network adapters. Furthermore, we can see where the configuration for each interface is coming from.

Similar result we can achieve by executing **sudo wicked ifstatus all**. If we want information for a specific adapter, we can substitute the word **all** with the adapter's name - **eth0**.

For compatibility reasons a valid and preferred way of managing configurations is to use the **ifcfg-xxx** files in **/etc/sysconfig/network**.

Let's examine the contents of the configuration file for adapter that has a valid IP address (**eth2**):

```
lsauser@jupiter:~> cat /etc/sysconfig/network/ifcfg-eth2
BOOTPROTO='dhcp'
BROADCAST=''
ETHTOOL_OPTIONS=''
IPADDR=''
MTU=''
NAME=''
NETMASK=''
NETWORK=''
REMOTE_IPADDR=''
STARTMODE='auto'
DHCLIENT_SET_DEFAULT_ROUTE='yes'
```

Let's create a copy of the file for our other network adapter (**eth1**) under different name if haven't created one earlier:

```
lsauser@jupiter:~> sudo cp /etc/sysconfig/network/ifcfg-eth2 /etc/sysconfig/network/ifcfg-eth1
```

We are going to use the new file to configure our network adapter #2 (**eth1**). Let's open it for editing (set the **bold** lines):

```
lsauser@jupiter:~> sudo vi /etc/sysconfig/network/ifcfg-eth1
BOOTPROTO='static'
BROADCAST=''
ETHTOOL_OPTIONS=''
```

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```
IPADDR='192.168.200.1/24'
MTU=''
NAME=''
NETMASK=''
NETWORK=''
REMOTE_IPADDR=''
STARTMODE='auto'
DHCLIENT_SET_DEFAULT_ROUTE='no'
```

Save and close the file (**:wq**).

If yours is with fewer lines, you may execute **man 5 ifcfg** to learn more about the possible options and values.

Next, we must restart the network service and check that everything is set as expected:

```
lsauser@jupiter:~> sudo systemctl restart network
lsauser@jupiter:~> ip a s eth1
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:4c:d3:b4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.200.1/24 brd 192.168.200.255 scope global eth1
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe4c:d3b4/64 scope link
       valid_lft forever preferred_lft forever
lsauser@jupiter:~> ip r
default via 10.0.2.2 dev eth2 proto dhcp
10.0.2.0/24 dev eth2 proto kernel scope link src 10.0.2.15
192.168.200.0/24 dev eth1 proto kernel scope link src 192.168.200.1
lsauser@jupiter:~>
```

Now, we are ready to move forward.

Network Manager

When using **Network Manager**, we can follow the same techniques as described in the corresponding section of the Practice for **AlmaLinux**.

Part 2: Software and Services

RPM

Let's first install a local package. In order to achieve this, we must download it first. Of course, we should get the right package. You can check by visiting https://zahariev.pro/linux/hello-lsa

On **AMD64**-based hardware, execute:

```
lsauser@jupiter:~> curl -O https://zahariev.pro/linux/hello-lsa/releases/hello-lsa-1.0-1.el9.x86_64.rpm
```

On **ARM64**-based hardware, execute:

```
lsauser@jupiter:~> curl -O https://zahariev.pro/linux/hello-lsa/releases/hello-lsa-1.0-1.el9.aarch64.rpm
```

List what we have so far. It should display the downloaded package. For example, for **AMD64**, we should see:

```
lsauser@jupiter:~> ls -l hello-lsa-1.0-1.el9.x86_64.rpm
-rw-r--r-- 1 lsauser users 10899 Mar 27 15:00 hello-lsa-1.0-1.el9.x86_64.rpm
lsauser@jupiter:~>
```

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We can ask for detailed information with:

```
lsauser@jupiter:~> rpm -qip hello-lsa-1.0-1.el9.x86_64.rpm
Name        : hello-lsa
Version     : 1.0
Release     : 1.el9
Architecture: x86_64
Install Date: (not installed)
Group       : Unspecified
Size        : 16273
License     : GPLv3+
Signature   : (none)
Source RPM  : hello-lsa-1.0-1.el9.src.rpm
Build Date  : Thu Mar 27 13:51:13 2025
Build Host  : almalinux
URL         : https://www.zahariev.pro/linux/hello-lsa
Summary     : Hello LSA greeting utility
Description :
Simple utility to display the text Hello LSA.
lsauser@jupiter:~>
```

In a similar way, we can find all files that will be installed by the package:

```
lsauser@jupiter:~> rpm -qlp hello-lsa-1.0-1.el9.x86_64.rpm
/usr/bin/hello-lsa
...
lsauser@jupiter:~>
```

Now that we have all the information, we can proceed with package installation:

```
lsauser@jupiter:~> sudo rpm -ivh hello-lsa-1.0-1.el9.x86_64.rpm
Preparing...                          ################################# [100%]
Updating / installing...
   1: hello-lsa-1.0-1.el9             ################################# [100%]
lsauser@jupiter:~>
```

It seems that all went well. Let's test it:

```
lsauser@jupiter:~> hello-lsa
```

Once we are done, we can remove it with:

```
lsauser@jupiter:~> sudo rpm -e hello-lsa
```

Zypper

Let's continue with **zypper**. We can refresh the information about the installed repositories:

```
lsauser@jupiter:~> sudo zypper refresh
...
All repositories have been refreshed.
lsauser@jupiter:~>
```

Now we can force the update of all installed packages. If there are any updates ready, they will be installed:

```
lsauser@jupiter:~> sudo zypper update -y
```

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```
...
lsauser@jupiter:~>
```

We can skip the **-y** option in order to be able to check what will be updated and to decide shall we continue or not.

We can use the shorter alias **up** instead.

After an upgrade, there could be running programs that use or refer to files that have been changed or removed. You can execute a command to check for those, and eventually restart either the processes or the system:

```
lsauser@jupiter:~> sudo zypper ps -s
The following running processes use deleted files:

PID   | PPID | UID   | User       | Command                  | Service
-----+------+------+------------+--------------------------+----------------
336   | 1    | 0     | root       | lvmetad                  | lvm2-lvmetad
...
```

Let's search for a tool command line web browser. We will search not only in the name, but in the description also:

```
lsauser@jupiter:~> zypper search -d browser
Loading repository data...
Reading installed packages...
S  | Name                                     | Summary                     | Type
---+------------------------------------------+-----------------------------+---------
...
   | lynx                                     | A Text-Based WWW Browser    | package
...
lsauser@jupiter:~>
```

Now that we know the name of the package, we can continue with the installation:

```
lsauser@jupiter:~> sudo zypper install lynx
...
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 2 NEW packages are going to be installed:
  lynx xli

2 new packages to install.
Overall download size: 1,7 MiB. Already cached: 0 B. After the operation, additional 7,7 MiB will be
used.
Continue? [y/n/...? shows all options] (y): y
...
lsauser@jupiter:~>
```

We can test our new browser by executing:

```
lsauser@jupiter:~> lynx cnn.com
```

We must confirm that we accept cookies with **A** for always. Once the page is loaded, we can explore a bit, and then quit with the **q** key.

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It is time to check what repositories we have installed and enabled on our system:

```
lsauser@jupiter:~> zypper repos --show-enabled-only
Repository priorities are without effect. All enabled repositories share the same priority.

# | Alias                                | Name                           | Enabled | GPG Check | Refresh
--+--------------------------------------+--------------------------------+---------+-----------+--------
1 | https-download.opensuse.org-c91fb5cd | https://download.opensuse.o-> | Yes     | (r ) Yes  | Yes
4 | openSUSE:repo-openh264               | repo-openh264 (16.0)           | Yes     | (r ) Yes  | Yes
5 | openSUSE:repo-oss                    | repo-oss (16.0)                | Yes     | (r ) Yes  | Yes
lsauser@jupiter:~>
```

These are the default repositories on a typical **openSUSE** system. Of course, we can end up in a situation in which we must add a repository. Different repositories can be installed by following different approaches. There are multiple ways:

- using the **yast2** tool (in versions before Leap 16.0)
- using the **zypper** tool
- creating or copying a **repo** file in **/etc/zypp/repos.d/** and etc.

Other **openSUSE** repositories can be found on the following places https://en.opensuse.org/Package_repositories and https://en.opensuse.org/Additional_package_repositories

We are ready to continue with the next step.

Before we close this section, we can install the missing legacy networking tools. First, we need to know how the package is named. We can use a special tool for this purpose:

```
lsauser@jupiter:~> cnf ifconfig


The program 'ifconfig' can be found in following packages:
  * net-tools-deprecated [ path: /bin/ifconfig, repository: zypp (repo-oss) ]
  * net-tools-deprecated [ path: /usr/bin/ifconfig, repository: zypp (repo-oss) ]

Try installing with:
    sudo zypper install net-tools-deprecated

lsauser@jupiter:~>
```

Now, we know what package to install in order to have it. We can do it now, or test another approach for the same:

```
lsauser@jupiter:~> zypper search -d ifconfig
Loading repository data...
Reading installed packages...

S  | Name                 | Summary                                      | Type
---+----------------------+----------------------------------------------+--------
   | gnome-nettool        | GNOME Interface for Various Networking Tools | package
   | moreutils            | Additional Unix Utilities                    | package
i  | net-tools            | Important Programs for Networking            | package
   | net-tools-deprecated | Deprecated Networking Utilities              | package
```

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```
   | python313-sh         | Python subprocess interface                  | package
lsauser@jupiter:~>
```

We can ask for detailed information:

```
lsauser@jupiter:~> zypper info net-tools-deprecated
Loading repository data...
Reading installed packages...

Information for package net-tools-deprecated:
----------------------------------------------
Repository     : https://download.opensuse.org/distribution/leap/16.0/repo/oss/x86_64
Name           : net-tools-deprecated
Version        : 2.10-160000.2.2
Arch           : x86_64
Vendor         : SUSE LLC <https://www.suse.com/>
Installed Size : 456.3 KiB
Installed      : No
Status         : not installed
Source package : net-tools-2.10-160000.2.2.src
Upstream URL   : https://sourceforge.net/projects/net-tools/
Summary        : Deprecated Networking Utilities
Description    :
    This package contains the deprecated network utilities arp, ifconfig, netstat and route,
    which have been replaced by tools from the iproute2 package:
      * arp -> ip [-r] neigh
      * ifconfig -> ip a
      * netstat -> ss [-r]
      * route -> ip r
lsauser@jupiter:~>
```

As usual, there are many ways to achieve something. We can install the package and experiment with those tools or with their replacements as suggested by the last few lines above.

Last but not least, should we want to see which packages are installed from particular repository, we can execute:

```
lsauser@jupiter:~> zypper search -i -r 5
...
S | Name                        | Summary                                                  | Type
--+-----------------------------+----------------------------------------------------------+--------
i | gstreamer-plugins-bad       | GStreamer Streaming-Media Framework Plug-Ins             | package
i | gstreamer-plugins-bad-lang  | Translations for package gstreamer-plugins-bad           | package
...
lsauser@jupiter:~>
```

By changing the above command a bit, we can see all the available packages (and patterns) in a repository:

```
lsauser@jupiter:~> zypper search -r 5
...
```

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```
S | Name                        | Summary                                                    | Type
--+-----------------------------+------------------------------------------------------------+--------
  | 32bit                       | 32-Bit Runtime Environment                                 | pattern
v | apparmor                    | AppArmor                                                   | pattern
...
lsauser@jupiter:~>
```

Dependencies Exploration

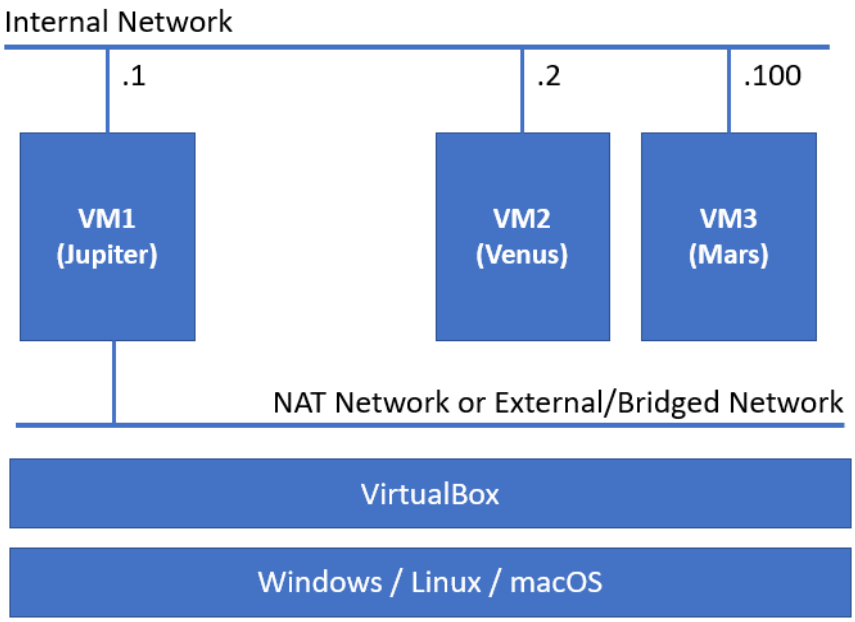
We can ask about the dependencies between binary and all shared libraries that it may need. This way, we can find any unmet dependencies. For this, we can use the **ldd** command:

```
lsauser@jupiter:~> ldd /bin/ls
        linux-vdso.so.1 (0x00007ffea0700000)
        libselinux.so.1 => /lib64/libselinux.so.1 (0x00007f7ff12f4000)
        libcap.so.2 => /usr/lib64/libcap.so.2 (0x00007f7ff10ef000)
        libc.so.6 => /lib64/libc.so.6 (0x00007f7ff0d35000)
        libpcre.so.1 => /usr/lib64/libpcre.so.1 (0x00007f7ff0aa8000)
        libdl.so.2 => /lib64/libdl.so.2 (0x00007f7ff08a4000)
        /lib64/ld-linux-x86-64.so.2 (0x00007f7ff173f000)
lsauser@jupiter:~>
```

Part 3: Network Services

Goal

Our goal is to create a setup like the one below by applying what we learned so far:



We can assume that we have:

- Server – **openSUSE VM #1** – machine with **two network adapters** (we can reuse the one prepared during part 2) that will act as a **router**. In addition, it will run the **DHCP service**. Until the end of the practice, it will be referred to as **jupiter**;

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- Stations – **openSUSE VM #2** and **#3** – those are standard VMs with just **one network adapter**. They will be connected only to the **internal network** (as shown in the picture). Until the end of the document, they will be referred as **venus** and **mars**;

If you are short on resources, you can use just the server and one station. You can also lower the RAM of the VMs to **512 MB**.

Preparation

As we will continue with the machine used in the previous two parts, we do not have to do anything to prepare it.

We must focus on the workstations. We must import one or more, but do not forget during the import process to activate the option for **MAC** re-initialization. In addition, as a post-import step, we must switch the network adapter of each workstation to **Internal Network** mode and select the name used for the second card of the server.

Depending on the network stack in use, we may have to take care of different things.

(Station 1) Configure with Static IP Address

Start station 1, login as a regular user and execute:

```
lsauser@opensuse:~> sudo hostnamectl set-hostname venus.lsa.lab
lsauser@opensuse:~> sudo hostnamectl set-hostname --pretty 'Venus Station'
lsauser@opensuse:~> hostnamectl
   Static hostname: venus.lsa.lab
   Pretty hostname: Venus Station
...
lsauser@opensuse:~>
```

Next, we must check which network platform is running our network and act accordingly.

We must set our adapter to be with static IP **192.168.200.2/24** with gateway set to **192.168.200.1** (in case of **wicked** create or modify the file **/etc/sysconfig/network/routes** (or **/etc/sysconfig/network/ifroute-eth1**) and make sure that the following line exists **default 192.168.200.1 - eth1**) and DNS set to **8.8.8.8** (in case of **wicked** modify the file **/etc/sysconfig/network/config** by altering the parameter **NETCONFIG_DNS_STATIC_SERVERS**).

Depending on the network stack, we can use either **nmtui** or **yast2** to ease our job.

Once done, we can restart the machine.

(Station 2) Configure with Dynamic IP Address

Start station 2, login as a regular user and execute:

```
lsauser@opensuse:~> sudo hostnamectl set-hostname mars.lsa.lab
lsauser@opensuse:~> sudo hostnamectl set-hostname --pretty 'Mars Station'
lsauser@opensuse:~> hostnamectl
   Static hostname: mars.lsa.lab
   Pretty hostname: Mars Station
...
lsauser@opensuse:~>
```

By default, the network adapter is set to DHCP, so we don't need to change anything.

We can restart the machine.

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(Server) Install and Connect Via SSH

Start the server, if not running, and log in as a regular user.

In case, due to some reason, we do not have an **SSH** installed and working on the server, we can do it by:

```
lsauser@jupiter:~> sudo zypper install -y openssh
...
lsauser@jupiter:~> sudo systemctl start sshd
lsauser@jupiter:~> sudo systemctl enable sshd
lsauser@jupiter:~> sudo firewall-cmd --add-service=ssh --permanent
lsauser@jupiter:~> sudo firewall-cmd --reload
```

Assuming that we have a working **SSH**, we can connect to our server and start having a better experience. We will be able to copy and paste commands and exchange files between the host and the **VM**.

If our host is **Windows** based, we have two options:

- If we have recent and updated **Windows 10** or newer, we can enable the **OpenSSH Client** optional feature;
- No matter which version of **Windows** we use, we can install one of the popular tools, for example **PuTTY** for handling **SSH** connections, and perhaps **WinSCP** for moving files between our host and the **VMs**;

Now on, we will assume that we have **OpenSSH Client** installed. This will make all steps that follow the same for **Windows**, **Linux**, or **macOS** host.

If our **VM's** network adapter is set to **NAT**, we must create a forwarding rule in our virtualization solution. For **VirtualBox**, we must do:

- Open the **VM** settings;
- Go to **Network**;
- Select the **NAT** adapter and open **Advanced** section;
- Click on **Port Forwarding**;
- Create a rule – for **host port** set for example **20022**, and for VM port – **22**. Then click **OK**
- Close the **Setting** window by clicking **OK**

Now, we can open a terminal window and enter the following to establish a session:

```
ssh -p 20022 lsauser@localhost
```

We can copy files to the **VM** by executing:

```
scp -P 20022 myfile-local.txt lsauser@localhost:/tmp/myfile-remote.txt
```

Or we can download files from the **VM** to the host (save the file in the current folder on the host) with:

```
scp -P 20022 lsauser@localhost:/some-folder/myfile-remote.txt .
```

Please note, that the above paths and filenames are arbitrary, you must substitute them with ones appropriate in your situation.

If our **VM's** network adapter is set to **Bridge**, we can omit rule creation and both **-p 20022** and **-P 20022** and change the **localhost** to the **VM's IP** address (which we can get with the **ip** command) in the above commands.

Install and Configure DHCP

The package needed is named **dhcp-server** and we can install it with:

```
lsauser@jupiter:~> sudo zypper install -y dhcp-server
...
```

Our first task is to configure the network interface (set it to listen on the one with the static IP address):

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```
lsauser@jupiter:~> sudo vi /etc/sysconfig/dhcpd
...
# Examples: DHCPD_INTERFACE="eth0 eth1 eth2"
#           DHCPD_INTERFACE="ANY"
#
DHCPD_INTERFACE="enp0s8"
...
```

Then we can modify the configuration of the service:

```
lsauser@jupiter:~> sudo vi /etc/dhcpd.conf
...
```

Enter in edit/insert mode and type the following (let's say just after the header):

```
option domain-name "lsa.lab";
option domain-name-servers 8.8.8.8;

subnet 192.168.200.0 netmask 255.255.255.0 {
range 192.168.200.100 192.168.200.120;
option routers 192.168.200.1;
option broadcast-address 192.168.200.255;
default-lease-time 600;
max-lease-time 7200;
}
```

Save the file and quit the editor (if using **vi**, then press **Esc**, and type **:wq** and hit **Enter**).

Now, it is a good time to test if we have a good configuration file. Execute:

```
lsauser@jupiter:~> sudo dhcpd -t
...
```

If we see the word **error** in the output, we must go back and correct the file.

With this the process of base DHCP configuration is over. We must start and enable the service:

```
lsauser@jupiter:~> sudo systemctl start dhcpd
lsauser@jupiter:~> sudo systemctl enable dhcpd
Created symlink /etc/systemd/system/dhcp-server.service → /usr/lib/systemd/system/dhcpd.service.
Created symlink /etc/systemd/system/multi-user.target.wants/dhcpd.service →
/usr/lib/systemd/system/dhcpd.service.
lsauser@jupiter:~> systemctl status dhcpd
...
```

Now our workstations will receive their IP addresses. We can log to the one (**Mars**) that is set to receive its IP address via DHCP and check if it got new address:

```
lsauser@mars:~> ip a
...
```

We can return on the server and check the leases database with:

```
lsauser@jupiter:~> cat /var/lib/dhcp/db/dhcpd.leases
...
```

We should have a working **DHCP** service, but our stations cannot access the Internet yet.

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(Server) Manage the Firewall

Return to the server (**Jupiter**) and check the status and current configuration of the firewall:

```
lsauser@jupiter:~> sudo firewall-cmd --get-active-zones
docker
  interfaces: docker0
public (default)
  interfaces: enp0s3 enp0s8
lsauser@jupiter:~> sudo firewall-cmd --get-zones
block dmz docker drop external home internal nm-shared public trusted work
lsauser@jupiter:~> sudo firewall-cmd --get-zone-of-interface=enp0s3
public
lsauser@jupiter:~> sudo firewall-cmd --get-zone-of-interface=enp0s8
public
lsauser@jupiter:~>
```

Now, let's set each interface in separate zone:

```
lsauser@jupiter:~> sudo firewall-cmd --zone=external --change-interface=enp0s3 --permanent
success
lsauser@jupiter:~> sudo firewall-cmd --zone=trusted --change-interface=enp0s8 --permanent
success
lsauser@jupiter:~> sudo firewall-cmd --reload
success
lsauser@jupiter:~> sudo firewall-cmd --get-active-zones
docker
  interfaces: docker0
trusted
  interfaces: enp0s8
public (default)
  interfaces: enp0s
lsauser@jupiter:~>
```

If the **enp0s3** is still not shown as in the **external** zone (as above), then restart the connection

```
lsauser@jupiter:~> sudo nmcli conn down enp0s3; sudo nmcli conn up enp0s3
```

Now it should be in the right zone.

(Stations) Adjust Network Configuration

Now, we can go to one of the workstations and check for Internet connectivity:

```
lsauser@venus:~> ping -c 1 softuni.bg
PING softuni.bg (217.174.159.195) 56(84) bytes of data.
64 bytes from softuni.bg (217.174.159.195): icmp_seq=1 ttl=63 time=1.20 ms

--- softuni.bg ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.205/1.205/1.205/0.000 ms
lsauser@venus:~>
```

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If everything is working as expected, we could update the stations.

Let's modify the settings of the **sshd** service running on **Mars** host. We will change the port to **50022** for example:

```
lsauser@mars:~> sudo vi /etc/ssh/sshd_config.d/alternative-port.conf
Port 50022
...
lsauser@mars:~> sudo systemctl restart sshd
lsauser@mars:~> systemctl status sshd
...
lsauser@mars:~> sudo firewall-cmd --add-port=50022/tcp --permanent
lsauser@mars:~> sudo firewall-cmd --reload
```

Now, return to the **Jupiter** server and try to connect to **Mars** via **ssh**:

```
lsauser@jupiter:~> ssh -p 50022 lsauser@192.168.200.100
The authenticity of host '[192.168.200.100]:50022 ([192.168.200.100]:50022)' can't be established.
ECDSA key fingerprint is SHA256:wQ+AJGr5G24fMloLq65Wwn77STLUMDCyc8E9nyaNZRE.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '[192.168.200.100]:50022' (ECDSA) to the list of known hosts.
Password:
Last login: Tue Sep 24 00:43:04 2019
Have a lot of fun...
lsauser@mars:~> hostnamectl
   Static hostname: mars.lsa.lab
Transient hostname: mars
         Icon name: computer-vm
           Chassis: vm
        Machine ID: 16333b0f57864155997c10082338fe24
           Boot ID: d851f57c83a944ad84f7a53049d9f10c
    Virtualization: oracle
  Operating System: openSUSE Leap 15.1
       CPE OS Name: cpe:/o:opensuse:leap:15.1
            Kernel: Linux 4.12.14-lp151.27-default
      Architecture: x86-64
lsauser@mars:~> exit
logout
Connection to 192.168.200.100 closed.
lsauser@jupiter:~>
```

Of course, we can revert the changes in the **sshd** configuration if we want to.

(Server) SSH Keys *

We can use key instead of the traditional password. Let's try it.

Return on the server amd generate a pair of private and public key (hit the **Enter** key four times):

```
lsauser@jupiter:~> ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/lsauser/.ssh/id_rsa):
```

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```
Created directory '/home/lsauser/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/lsauser/.ssh/id_rsa
Your public key has been saved in /home/lsauser/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:m+6fKWLNGcPc71FLu1/rZwA7tR2UeauN67cfF+YyyNE lsauser@jupiter
The key's randomart image is:
+---[RSA 3072]----+
|                o|
|               +.|
|              . o|
|            o .o |
|       oS. . E*=.|
|        =oo ===+o|
|       oo+ +.+++o|
|      o.=  o.oooB|
|     . ooo+.o.+**|
+----[SHA256]-----+
lsauser@jupiter:~>
```

By default this will create a pair of keys using the RSA algorythm which are not password protected and will store them in the .ssh subforlder of the home folder of our user:

```
lsauser@jupiter:~> ls -al ~/.ssh
total 8
drwx------ 2 lsauser users   38 Sep 23 16:04 .
drwxr-xr-x 8 lsauser users  184 Sep 23 16:04 ..
-rw------- 1 lsauser users 2602 Sep 23 16:04 id_rsa
-rw-r--r-- 1 lsauser users  569 Sep 23 16:04 id_rsa.pub
lsauser@jupiter:~>
```

Should we want to change this behavior, we can add the appropriate options on the command line.

Now, how do we use these keys? We must copy the public key to all stations that we want to use it on.

For example, let's copy it on the Venus machine by executing:

```
lsauser@jupiter:~> ssh-copy-id 192.168.200.2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/lsauser/.ssh/id_rsa.pub"
The authenticity of host '192.168.200.2 (192.168.200.2)' can't be established.
ED25519 key fingerprint is SHA256:2n9I5AgmewInhFyLfhsvMSGHgETgg6i62dm5lyhTR8E.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already
installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install
the new keys
lsauser@192.168.200.2's password:
```

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```
Number of key(s) added: 1

Now try logging into the machine, with:   "ssh '192.168.200.2'"
and check to make sure that only the key(s) you wanted were added.

lsauser@jupiter:~>
```

You will be asked to enter the password for the lsauser on the other machine (in our case it is existing there and has the same password).

Now, we can try and do the SSH session just like:

```
lsauser@jupiter:~> ssh 192.168.200.2
Last login: Fri Sep 23 14:04:13 2022
lsauser@venus:~>
```

And bam, without being asked for a password we managed to establish a session.

Close the session and return to the **Jupiter** machine.

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